Assignment_Determinant

1.A matrix $M \in \mathrm{M}_{n\times n}(R)$ is called **skew-symmetric** if $M^t = -M$.
Prove that if M is skew-symmetric and n is odd, then M is not invertible.
What happens if n is even?

2.A matrix $Q \in \mathrm{M}_{n\times n}(R)$ is called **orthogonal** if $QQ^t = I$. Prove that if Q is orthogonal, then $\det(Q) = \pm 1$.

3.Let C be the adjugate matrix of $A \in \mathrm{M}_{n\times n}(R)$. Prove the following statements.
(a) $\det(C) = [\det(A)]n-1$.
(b) C^t is the adjugate matrix of A^t.
(c) If A is an invertible upper triangular matrix, then C and A^{-1} are both upper triangular matrices.

4.Let V b e the vector space of nxn matrices. Let B be a fixed element of V and let T_B be the linear operator on V defined by $T_B(A) = AB - BA$.
Show that $\det T_B = O$.

5.If V is the vector space of nxn matrices and B is a fixed nxn matrix, let L_B and R_B be the linear operators on V defined by $L_B(A) = BA$ and $R_B(A) = AB$. Show that
(a) $\det L_B = (\det B)^n$;
(b) $\det R_B = (\det B)^n$.

6. Let A, B, C, D be commuting nxn matrices. Show that the determinant of the 2nx2n matrix $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$ is det(AD-BC).

7.Suppose A is an mxn matrix and B is an nxm matrix.

Show that $\det(I_m + AB) = \det(I_n + BA)$.

8.Suppose $A=(a_{ij})$ such that each entry is a differentiable function of t near t=0. Suppose A is an invertible matrix at t=0. Find $\frac{d}{dt}\log\det A$.